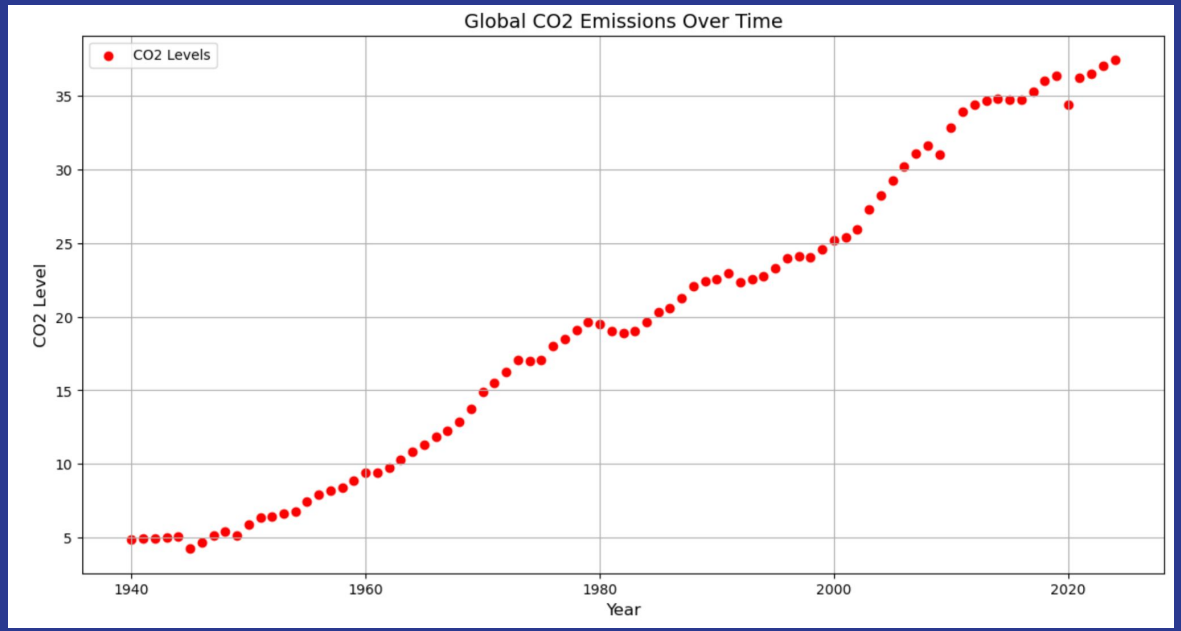
Analysis of CO_2 Emission Trends

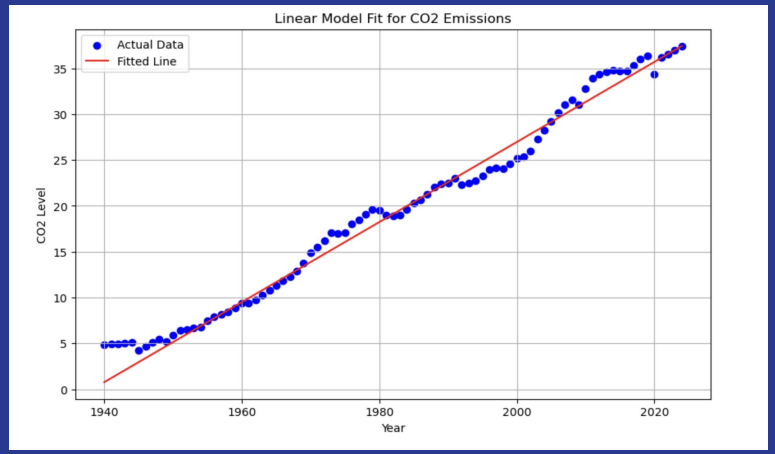
Objective: To examine historical global CO_2 emissions trends by applying three distinct computational models (Linear, Polynomial, Exponential), using gradient descent optimization to fit the data.The analysis aims to uncover key patterns, evaluate model performance, and provide insights that inform us about this topics

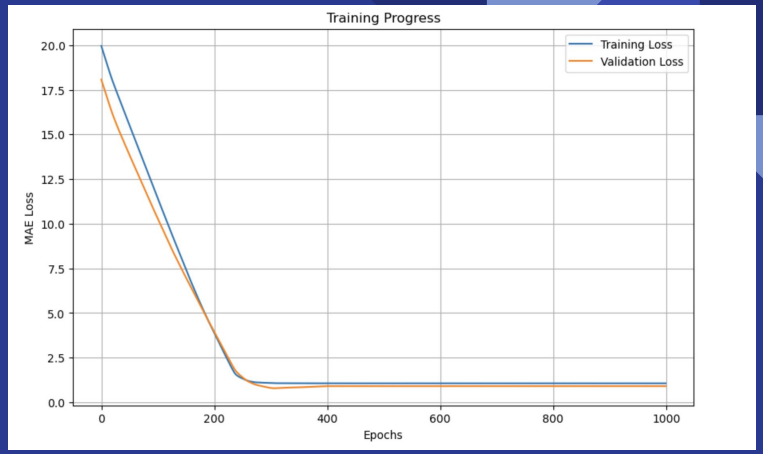


The image aside represents annual global CO22 emissions levels from 1940 to 2024.

From 1940 to the mid-20th century, the increase in emissions was gradual, coinciding with slower industrial growth during and after World War II. However, post-1960, there is a noticeable acceleration in emissions, reflect globalization and perhap population growth

Linear Model





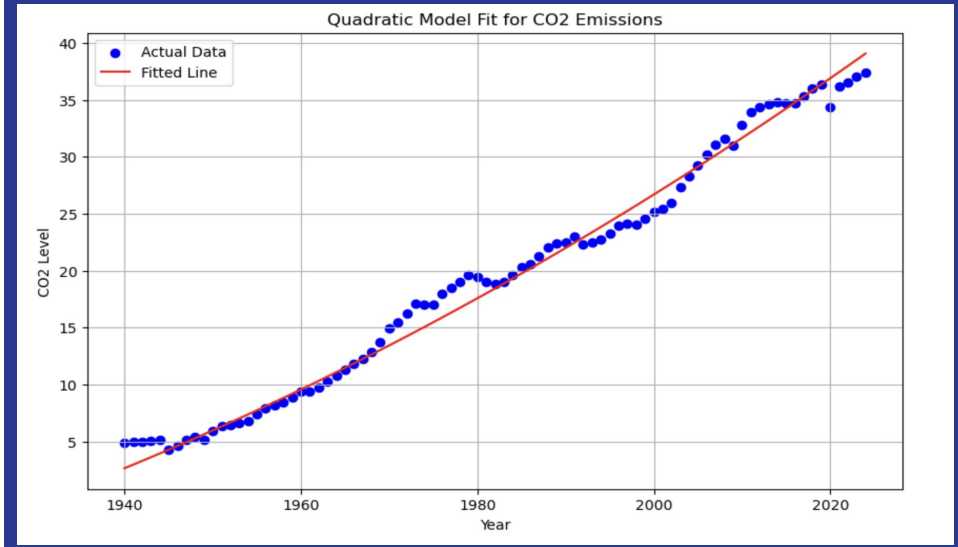
The fitted model(Image of the left), represented by the red line, successfully captures the general upward trend of CO22 emissions over time, as seen in the blue data points. However, the linear model fails to capture the non-linear acceleration in emissions, particularly after 1960. However, as time goes on (1980-2020), the actual data starts to rise faster than the model predicts. This shows that a simple linear model is not enough to capture the accelerating growth in CO22 emissions caused by factors like industrialization and global energy demand.

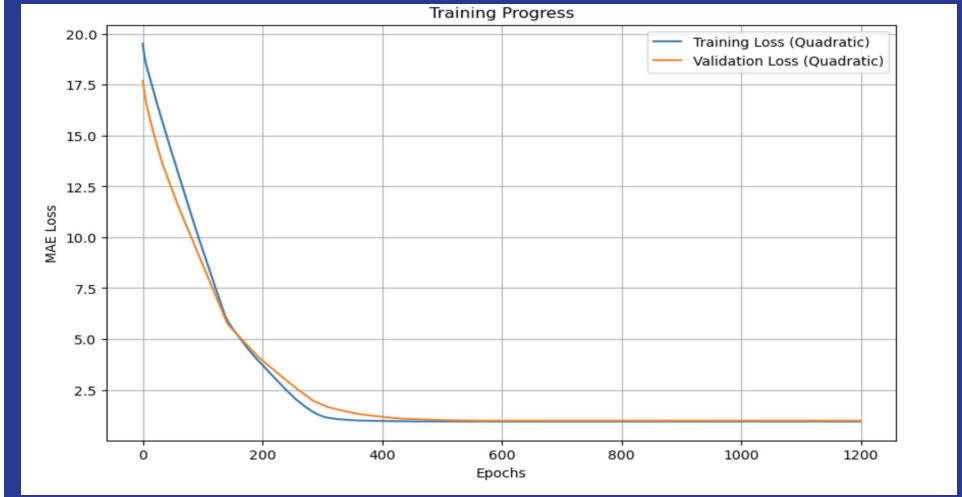
Looking at the training progress graph, the model quickly learns the linear pattern, with the loss (or error) dropping sharply in the first 200 epochs. After that, the loss levels out, meaning the model has reached its best performance. However, the error remains relatively high because once again linear model cannot fully explain the complexity of the data.

Quadratic Model

Unlike the linear model, the quadratic model captures the accelerating growth in emissions, particularly after 1960. In earlier years (1940-1960), the model aligns well with the steady growth pattern, and in later years (1980-2020), it adjusts to the increasing rate of emissions. This illustrates quadratic model's ability to handle non-linear trends in the data, making it a much better fit for capturing the historical progression of CO22 emissions.

The training progress graph further validates the effectiveness of the quadratic model. Both the training and validation losses decrease rapidly during the first 300 epochs and continue to decline until they converge around an MAE of approximately 0.5. This significantly lower error compared to the linear model indicates that the quadratic model is better suited for this dataset

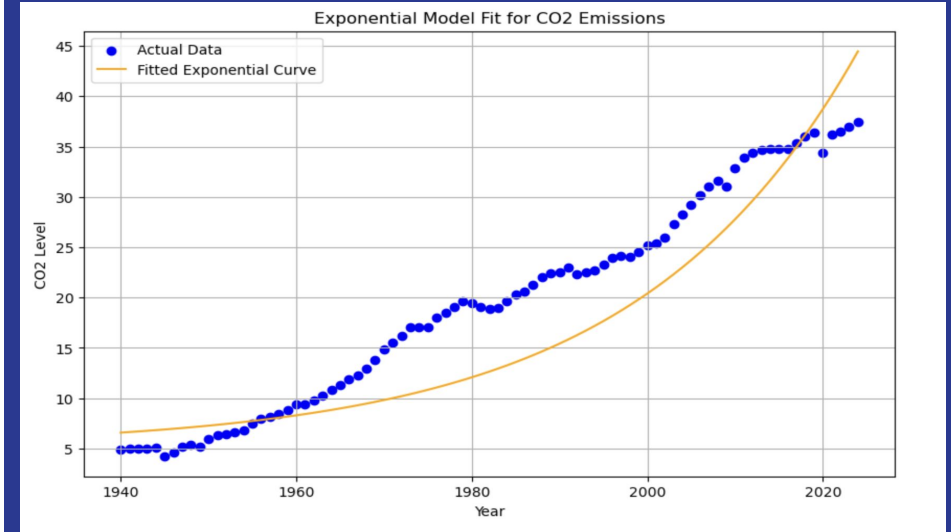


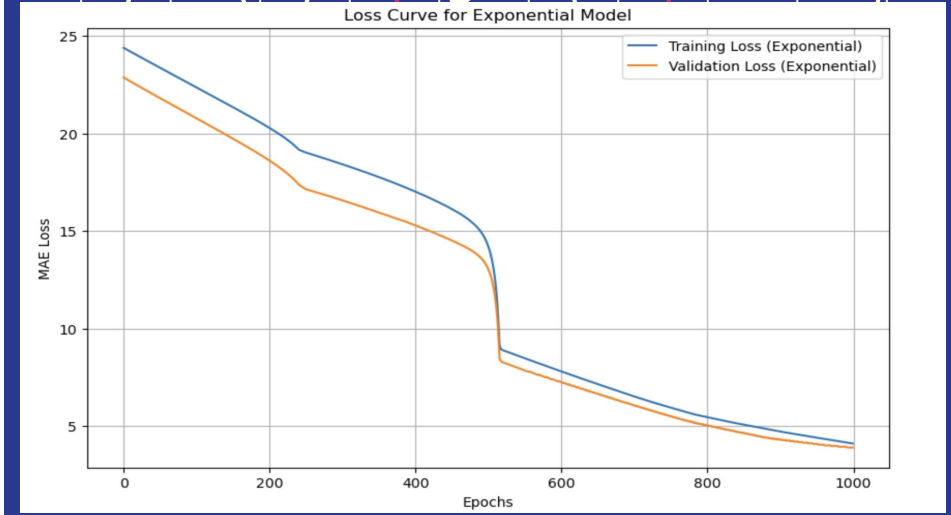


Exponential Model

While the model aligns with the general trend of emissions increasing at an accelerating rate, it tends to overestimate the CO22 levels in the later years (post-2000). This overestimation suggests that while emissions have grown significantly, the increase is not strictly exponential across the entire timeframe. In earlier periods (1940-1980), the model follows the data more closely, but its performance diminishes as the data becomes less uniformly exponential.

The training progress graph shows that both the training and validation losses decrease steadily and converge around an MAE of about 2.0. The training and validation losses are close, meaning the model generalizes well to new data. However, the higher error compared to the quadratic model shows that the exponential model struggles to fit the data as precisely, especially when the growth rate slows down or changes.





Conclusion

In conclusion, the linear model provides a basic understanding but oversimplifies the data. The quadratic model offers the best fit, capturing both steady growth and acceleration in emissions over time. The exponential model effectively shows rapid growth but struggles with accuracy in later years. Overall, the quadratic model is the most reliable for understanding historical trends and informing future predictions, emphasizing the need for nuanced approaches to studying emissions data